

James R. Hermus

CONTACT INFORMATION	Idiap Research Institute Centre du Parc, Rue Marconi 19, 1920 Martigny, Switzerland	<i>E-mail:</i> james.hermus@idiap.ch <i>Website:</i> jameshermus.github.io <i>ORCID:</i> 0000-0001-8926-8860
RESEARCH INTERESTS	Robotic Manipulation, Physical Interaction, Human Motor Control, Learning from Demonstration, System Identification, and Solid Sampling.	
EDUCATION	Massachusetts Institute of Technology , Cambridge, Massachusetts Ph.D., <i>Mechanical Engineering</i> September 2022 • <i>Thesis:</i> “A Dynamic Primitives Hypothesis: A Descriptive Model of Human Physical Interaction”, <i>Defense:</i> 9/9/22, <i>Committee:</i> Neville Hogan, Alberto Rodriguez, Harry Asada • <i>Fellowship in Graduate Coaching, Kaufman Teaching Certificate</i> • Course work: Controls, Dynamics, System ID, Machine Learning, Robotic Manipulation Massachusetts Institute of Technology , Cambridge, Massachusetts S.M., <i>Mechanical Engineering</i> June 2018 University of Wisconsin-Madison , Madison, Wisconsin B.S., <i>Biomedical Engineering</i> May 2016 • <i>Honors in Research, Biology in Engineering Certificate</i>	
RESEARCH EXPERIENCE	SNSF Ambizione Fellow and Principal Investigator September 2026 - Present <i>Robot Learning & Interaction Group</i> Idiap Research Institute Lab Director: Sylvain Calinon PhD Postdoctoral Researcher October 2024 - August 2026 <i>Robot Learning & Interaction Group and Human-centered Robotics and AI Group</i> Idiap Research Institute Lab Director: Sylvain Calinon PhD; Co-advisor: Emmanuel Senft PhD Postdoctoral Researcher February 2023 - May 2024 <i>The Learning Algorithms and Systems (LASA) Laboratory</i> École Polytechnique Fédérale de Lausanne (EPFL) Lab Director: Professor Aude Billard Postdoctoral Researcher September 2022 - January 2023 Graduate Research Assistant October 2016 - September 2022 <i>The Eric P. and Evelyn E. Newman Laboratory for Biomechanics and Human Rehabilitation</i> Massachusetts Institute of Technology Lab Director: Professor Neville Hogan Undergraduate Research Assistant May 2015 - August 2016 <i>Neuromuscular Biomechanics Lab</i> , University of Wisconsin-Madison Lab Director: Professor Darryl Thelen Undergraduate Research Assistant May 2013 - May 2015 <i>Department of Medical Physics</i> , University of Wisconsin-Madison Lab Directors: Professor Charles Mistretta and Professor Timothy P. Szczekutowicz	
GRANTS & FUNDING	Swiss National Science Foundation (SNSF) Ambizione Grant 2026 - 2030 <i>Confronting the Paradox of Human Motor Control: Leveraging Subtractive and Generative Models</i> Principal Investigator; 876,934 CHF (approx. 1.1M USD) ; Idiap Research Institute Grant No. 233246; awarded 2025	

	Contributions to EU Horizon 2020 Projects, EPFL <i>I.A.M.</i> (Impact-Aware Manipulation): led EPFL's work on dual-arm grasping and hitting <i>DARKO</i> : managed EPFL's work on robotic throwing	2023 - 2024
HONORS & AWARDS	Institute of Electrical and Electronics Engineers (IEEE) Travel grant for IROS Workshop - Member Support Program Massachusetts Institute of Technology de Florez Graduate Science Competition - 2nd Place Mechanical Engineering Research Exhibition 1st Place (2021), Honorable Mention (2019), Runner-up (2018) Assistive Technology Hackathon - 1st Place Technical Design Harrington Fellowship (35k USD) University of Wisconsin-Madison Hilldale Undergraduate/Faculty Research Fellowship (4k USD) Undergraduate Scholarships Dallas R. Lamont, Fred W. & Josephine Colbeck, Wisconsin Experience Accessibility, and University Book Store Academic Excellence Award Steuber Prize for Excellence in First-Year Writing	2023 2022 2018 - 2021 2017 2016 2014 2013 - 2015 2013
TEACHING EXPERIENCE	Lecturer (50% time) Course 2.151: Advanced System Dynamics and Control Rating: 6.5/7 Massachusetts Institute of Technology Kaufman Teaching Certificate Program During the completion of the KTCP course offered by the Teaching + Learning Lab at MIT, we covered several important topics including course design, teaching for belonging, outlining a unit of a course, engaging students and facilitating learning, planning for learning, microteaching, feedback and learning, and syllabus construction.	Fall Term 2022 Fall Term 2022
INTELLECTUAL PROPERTY	Meghan Huber, James Hermus , Gabrielle Enns, and Neville Hogan (2020). Variable Compression Body Anchor. Patent Number US 11,690,776 B2. Date of Patent July 4, 2023.	
PEER REVIEWED PUBLICATIONS	James Hermus , Michael Bombile, Jari van Steen, Elise Jeandupeux, Ahmed Zermane, Alessandro Melone, Mario Troebinger, Abdeldjallil Naceri, Claude Lacoursière, Stijn de Looijer, Sami Haddadin, Abderrahmane Kheddar, Alessandro Saccon, and Aude Billard. Impact-Aware Dual-Arm Manipulation: A Framework for Fast Package Handling. <i>IEEE Robotics and Automation Magazine</i> 2025 [Link] Chenguang Zhang, Federico Tessari, James Hermus , Himanshu Akolkar, Neville Hogan, and Andrew Schwartz. Tuning of Task-Relevant Stiffness in Multiple Directions. <i>Scientific Reports</i> 15(29916) 2025 [Link] Harshit Khurana, James Hermus , Maxime Gautier, and Aude Billard. Learning the Inverse Hitting Problem. <i>IEEE Robotics and Automation Letters (RA-L)</i> 10(5) 2025 [Link] Federico Tessari, James Hermus , Rika Sugimoto-Dimitrova, and Neville Hogan. Brownian Processes in Human Motor Control Support Descending Neural Velocity Commands. <i>Scientific Reports</i> 14(8341) 2024 [Link] James Hermus , Joseph Doeringer, Dagmar Sternad, and Neville Hogan. Dynamic Primitives in Constrained Action: Systematic Changes in the Zero-Force Trajectory. <i>Journal of Neurophysiology</i> 131(1) 2024 [Link] A. Michael West Jr., James Hermus , Meghan Huber, Pauline Maurice, Dagmar Sternad, and Neville Hogan. Dynamic Primitives Limit Human Force Regulation during Motion. <i>IEEE Robotics and Automation Letters</i> 7(2) 2022 [Link]	

James Hermus, Johannes Lachner, David Verdi, and Neville Hogan. Exploiting Redundancy to Facilitate Physical Interaction. *IEEE Transactions on Robotics* **38(1)** 2022 [\[Link\]](#)

James Hermus, Joseph Doeringer, Dagmar Sternad, and Neville Hogan. Separating Neural Influences from Peripheral Mechanics: The Speed-Curvature Relation in Mechanically Constrained Actions. *Journal of Neurophysiology* **123(5)** 2020 [\[Link\]](#)

Jack A. Martin, Scott C.E. Brandon, Emily M. Keuler, **James R. Hermus**, Alexander C. Ehlers, Daniel J. Segalman, Matthew S. Allen, and Darryl G. Thelen. Gauging Force by Tapping Tendons. *Nature Communications* **9(1)** 2018 [\[Link\]](#)

James Hermus and Timothy P. Szczykutowicz. Two-Dimensional Dynamic Fluid Bowtie Attenuators. *Journal of Medical Imaging (JMI)* **3(1)** 2016 [\[Link\]](#)

Timothy P. Szczykutowicz, **James Hermus**, Mark Geurts, and Jennifer Smilowitz. Realization of Fluence Field Modulated CT on a Clinical TomoTherapy Megavoltage CT System. *Physics in Medicine and Biology* **60(18)** 2015 [\[Link\]](#)

Timothy P. Szczykutowicz and **James Hermus**. Creation of an Atlas of Filter Positions for Fluence Field Modulated CT. *Medical Physics* **42(4)** 2015 [\[Link\]](#)

MANUSCRIPTS UNDER REVIEW

Jasper Tan, **James Hermus**, Emmanuel Senft, Sylvain Calinon, and Pascal Miéville. ARMS: an Automated Robotized human-Mimicking Solid sampling system. *Digital Discovery* 2026 (Under Review). Preprint on *ChemRxiv* [\[Link\]](#)

Adrian Prados, **James Hermus**, Ramon Barber, and Sylvain Calinon. Identifiable Decomposition of Submovements in Human Hand Trajectories. *Communications Biology* 2026 (Under Review). Preprint on *arXiv* [\[Link\]](#)

CONFERENCE PUBLICATIONS

James Hermus, Dagmar Sternad, Neville Hogan. (2020, November). Evidence for Dynamic Primitives in a Constrained Motion Task. *8th IEEE International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, New York, NY. [\[Link\]](#)

Timothy P. Szczykutowicz, **James Hermus**, Mark Geurts, and Jennifer Smilowitz. (2015, June). Intensity Modulated Imaging?: Clinical Workflow for Fluence Field Modulated CT On a TomoTherapy System. Presented at the *2015 Annual Meeting of the American Association of Physicists in Medicine (AAPM)*. TH-EF-BRB-6 [\[Link\]](#)

Timothy P. Szczykutowicz and **James Hermus**. (2015, March). Fluence Field Modulated CT on a Clinical TomoTherapy Radiation Therapy Machine. Oral presentation at the *2015 Annual SPIE Medical Imaging Conference*, Proc. 9412, Orlando, FL. [\[Link\]](#)

James Hermus, Charles A. Mistretta and Timothy P. Szczykutowicz. (2015, March). Scatter Correction of Vessel Dropout Behind Highly Attenuating Structures in 4D-DSA. Poster presentation at the *2015 Annual SPIE Medical Imaging Conference*, Proc. 9412, Orlando, FL. [\[Link\]](#)

Timothy P. Szczykutowicz and **James Hermus**. Fluid Dynamic Bowtie Attenuators. (2015, March). Oral presentation (**I presented the talk**) at the *2015 Annual SPIE Medical Imaging Conference*, Proc. 9412-31, Orlando, FL. [\[Link\]](#)

James Hermus, Cameron Hays, Michal Adamski, Hannah Lider, Jenny Westlund, Austin Scholp, John Webster and Bjoern Buehring. (2015, May). Posture Monitor for Vibration Exercise Training. Oral presentation at the *2015 IEEE Great Lakes Biomedical Conference*, Milwaukee, WI. [\[Link\]](#)

James Hermus, Timothy P. Szczykutowicz, Brian Davis, Erick L. Oberstar, Martin Wagner, Charles M. Strother, and Charles Mistretta. (2014, March). Quantitative Analysis of Artifacts in 4D DSA: the Relative Contributions of Beam Hardening and Scatter to Vessel Dropout Behind Highly Attenuating Structures. Poster presented at the *2014 Annual SPIE Medical Imaging Conference*, Proc. 9033, San Diego, CA. [\[Link\]](#)

- CONFERENCE PRESENTATIONS Adrian Prados, **James Hermus**, Ramon Barber, and Sylvain Calinon. (2026, February). Uncovering Motor Control Primitives: A Gradient-Based Optimization Framework for Robust Submovement Decomposition. Poster at *First Workshop on Assistive Robotics in Switzerland*. Zurich, Switzerland.
- Harshit Khurana, **James Hermus**, Aude Billard. (2024, September). Impact Aware Extended Kalman Filter. Extended Abstract presented at the 2024 *40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA@40)*, Abstract no. 378, Rotterdam, Netherlands.
- James Hermus**, Michael Bosongo Bombile, Jari J. van Steen, Elise Jeandupeux, Ahmed Zermane, Alessandro Melone, Mario Troebinger, Abdeldjalil Naceri, Claude Lacoursière, Stijn de Looijer, Sami Haddadin, Abderrahmane Kheddar, Alessandro Saccon, Aude Billard. (2024, September). Impact-aware dual-arm manipulation. Video Submission presented at the 2024 *40th Anniversary of the IEEE International Conference on Robotics and Automation (ICRA@40)*, Abstract no. 212, Rotterdam, Netherlands.
- Chenguang Zhang, Himanshu Akolkar, Federico Tessari, **James Hermus**, Neville Hogan, Andrew Schwartz. (2024, October). Neural coding of force and stiffness during a multi-directional ballistic-release task. Poster presented at the 2024 *Annual Conference of the Society for Neuroscience*, Abstract no. 6187, Chicago, IL.
- James Hermus**, Federico Tessari, Rika Sugimoto-Dimitrova, Neville Hogan. (2023, November). Velocity-level Planning in Human Neuro-motor Control: Behavioral Evidence Based on Brownian Processes. Poster presented at the 2023 *Annual Conference of the Society for Neuroscience*, Abstract no. 5698, Washington, D.C.
- Federico Tessari, Chenguang Zhang, Himanshu Akolkar, **James Hermus**, Neville Hogan, Andrew Schwartz. (2023, November). Direction-independent Stiffness Regulation in a Challenging Ballistic Release Tasks Highlights Human Neuro-Motor Performance Limitations. Poster presented at the 2023 *Annual Conference of the Society for Neuroscience*, Abstract no. 5627, Washington, D.C.
- Chenguang Zhang, Himanshu Akolkar, Federico Tessari, **James Hermus**, Neville Hogan, Andrew Schwartz. (2023, November). Direction-independent Impedance in Non-human Primates. Poster presented at the 2023 *Annual Conference of the Society for Neuroscience*, Abstract no. 5711, Washington, D.C.
- Himanshu Akolkar, Chenguang Zhang, Federico Tessari, **James Hermus**, Neville Hogan, Andrew Schwartz. (2023, November). Task-dependent Stiffness is independent of Movement Direction and Muscle Activation. Poster presented at the 2023 *Annual Conference of the Society for Neuroscience*, Abstract no. 5016, Washington, D.C.
- Chenguang Zhang, Himanshu Akolkar, Federico Tessari, **James Hermus**, Neville Hogan, Andrew Schwartz. (2022, November). Arm Impedance in Different Movement Directions. Poster presented at the 2022 *Annual Conference of the Society for Neuroscience*, Abstract no. 473.09, San Diego, CA.
- James Hermus**, Johannes Lachner, David Verdi, and Neville Hogan. (2022, May). Exploiting Redundancy to Facilitate Physical Interaction. Talk and poster at the 2022 *IEEE International Conference on Robotics and Automation*, Philadelphia, PA.
- Michael West, Meghan Huber, **James Hermus**, Pauline Maurice, Dagmar Sternad, and Neville Hogan. (2021, April). Humans Do Not Directly Control Force During Motion. Poster at the 2021 *Annual Conference of the Society for the Neural Control of Movement*, Online.
- James Hermus**, Dagmar Sternad, and Neville Hogan. (2020, October). Features of Free Motion Persist in Constrained Actions. Poster at the 2021 *IEEE International Conference on Intelligent Robots and Systems, Workshop on Learning Impedance Modulation for Physical Interaction*, Online.

Laura Schwendeman, **James Hermus**, Neville Hogan. (2020, October). A Frame-Based Approach to Submovement Decomposition. Interactive talk at the 2020 *Neuromatch Conference*, Online.

James Hermus, Dagmar Sternad, Neville Hogan. (2019, October). Dynamic Primitives Account for Human Constrained Motion. Poster presented at the 2019 *Annual Conference of the Society for Neuroscience*, Chicago, IL.

James Hermus, Dagmar Sternad, Neville Hogan. (2019, May). Features of Free Motion Persist in Constrained Actions. Poster presented at the 2019 *Annual IEEE International Conference on Robotics and Automation: Human movement science for physical human-robot collaboration workshop*, Montreal, Canada.

James Hermus, Joseph Doeringer, Dagmar Sternad, Neville Hogan. (2018, July). Physical Interaction with a Circular Constraint. Oral presentation at the 2018 *Annual International Society of Electrophysiology and Kinesiology*. Session Motor Control II, Dublin, Ireland.

Jack A. Martin, Emily M. Keuler, **James R Hermus**, Scott C.E. Brandon, Matthew S. Allen and Darryl G. Thelen. (2017, August). Achilles Tendon Wave Speed Tracks Joint Torque and Muscle Activity in Gait. **Received Best Oral Presentation Award** at the 2017 *Annual American Society for Biomechanics Conference*, Boulder, CO.

Jack A. Martin, Alexander C. Ehlers, **James R. Hermus**, Matthew S. Allen, Daniel J. Segalman and Darryl G. Thelen. (2017, June). Dynamic Imaging of Tendon Tissue Stress. **Received 3rd place award** for the conference paper. Oral presentation at the 2017 *Bi-annual Summer Biomechanics, Bioengineering and Biotransport Conference*, Tucson, AZ.

Jack A. Martin, Emily M. Keuler, **James R. Hermus**, Mikel R. Stiffler, Matthew S. Allen, and Darryl G. Thelen. (2016, August). Ultrasonic Imaging of In Vivo Achilles Tendon Stress During Walking. Presented at the 2016 *Annual American Society for Biomechanics Conference*, Raleigh, NC.

INVITED TALKS Confronting the Paradox of Human Motor Control. (2025, June). Invited speaker at the *Lorraine Research Laboratory in Computer Science and its Applications (LORIA)* in Nancy, France.

Robotic Manipulation from a Human Motor Control Perspective. (2024, January). Spotlight talk at the *IEEE Robotics and Automation Society Robotics Workshop* at EPFL, Lausanne, Switzerland.

Real Time Adaptive Systems for Human-Robot Collaboration. (2023, March). Invited speaker at the Industry 4.0 Workshop at *Swiss Robotics Innovation Booster*, Bern, Switzerland.

Quantifying Strengths and Weaknesses of Human Motor Control and Perception. (2022, May). Invited speaker at the *Workshop - Intelligent Control Methods and Machine Learning Algorithms for Human-Robot Interaction and Assistive Robotics* as part of the 2022 *IEEE International Conference on Robotics and Automation*, Philadelphia, PA.

Human Physical Interaction with a Circular Constraint. (2019, October). Invited speaker at the *UW-Madison Neuromuscular Biomechanics Lab*, Madison, WI.

UNIVERSITY SERVICE	Graduate Coach Fellow, MIT	2020-2022
	• Trained as peer/group-coach facilitator to lead discussions • Program based on International Coaching Federation standards and met weekly • Led group coaching sessions to empower the professional/personal development of graduate students	
	Makerworkshop, MIT	2017-2023
	• 3D printing Machine Master • Maintained Markforged, Stratasys, and Zortrax 3D printers for student use. • Trained students on 3D printing and consulted on research and personal design projects.	2018-2021
	MIT Outing Club, MIT	
	• Treasurer - Managed financials (greater than 200k USD)	2021

- Climbing leader and winter school leader **2017-2023**
 - Led climbing trips, taught technical rope skills, and shared a love for the outdoors.
- Advocates for Diverse Abilities (ADA), UW-Madison**
- **Co-founder and Vice President** **2012-2016**
 - Disability Awareness Student Organization
 - Received runner up for the Social Justice Advocacy Bucky Award **2015**
 - **Guest Speaker - Rehabilitation Psychology 300 Class** **2012-2016**
- Service Learning Trip – Dominican Republic, UW-Madison** **2013**
- Tutoring in Adaptive Technology - Kurzweil and Dragon Speaks, UW-Madison** **2014**

ROBOTICS
COMMUNITY
SERVICE

Workshop Organizer

- *IEEE Robotics and Automation Society Robotics Workshop at EPFL*. Lausanne, Switzerland. January 10, 2024.
- *Multilimb Coordination and Learning: an Interplay of Robotics and Human Neuroscience*. IEEE International Conference on Intelligent Robots and Systems (IROS). Detroit, MI. October 5, 2023. [\[Link\]](#)

Reviewer

- Scientific Reports
- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Rehabilitation Robotics (ICORR)
- Robotics: Science and Systems
- Hindawi Applied Bionics and Biomechanics

SUPERVISION
EXPERIENCE

Supervised or mentored 3 doctoral students (1 current), 1 visiting doctoral student, 6 Master's students, 7 undergraduate researchers, and 1 high school student at Idiap, EPFL, and MIT. Names and details available on request.

IN THE PRESS

"Idiap Researcher Awarded SNSF Ambizione Grant to Advance Human Motor Control and Robotics." *Idiap News*. October 17, 2025. [\[Link\]](#)

"Learning challenges shape a mechanical engineer's path." *MIT News*. February 12, 2023. [\[Link\]](#)

"Medical Meets Mechanical." *MechEConnects*. Winter 2018, Department Newsletter. [\[Link\]](#)

"McBurney Disability Resource Center." *Forward Motion*. Big Ten Network. June 12, 2015, Television. [\[Link\]](#)

REFERENCES

Available upon request.